Vansh Gupta

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CodeChef (max 1856) | Codeforces **Pupil** (max 1340) | New Delhi, India

Professional Summary

Third-year B.Tech IT student at IIIT Bhopal (CGPA: **8.49/10**). Six-month ML Intern at Moleculyst building Video LLM pipelines and agentic AI systems. Shipped production multi-agent LLM apps: a 3-agent paper explainer (1,100+ live items, 85 tests) and a 5-agent LangGraph pipeline on GCP. Hands-on with LLM reasoning, tool use, multi-turn dialogue, function calling, and structured output parsing across LLaMA 3.1, Gemini, GPT-4, Claude, and DeepSeek.

Education

IIIT Bhopal **Bhopal, Madhya Pradesh, India**
B.Tech in Information Technology (CS & Engineering Curriculum) | *CGPA: **8.49 / 10*** *Aug 2024 – May 2028*

- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra, Probability & Statistics, Discrete Mathematics, Operating Systems, Database Management Systems

Technical Skills

LLMs & Models: LLaMA 3.1, Gemini API, GPT-4, Claude, DeepSeek, Prompt Engineering, Function Calling, Multi-Turn Dialogue Design, Tool Use, Structured Output Parsing

Agentic Frameworks: LangChain, LangGraph, Multi-Agent Orchestration, OpenEnv, asyncio, JSON Data Formats, REST APIs, Reinforcement Learning Environments

ML / Deep Learning: PyTorch, TensorFlow, RAG, NLP, Transformers, SHAP, Scikit-learn, Computer Vision, VLMs

Languages & Engineering: Python, C++, SQL, FastAPI, Docker, Git, GCP (Cloud Run, Vertex AI), Supabase

Competitive Programming (C++): CodeChef (max **1856**) | **Codeforces Pupil** (max **1340**) — DP, graph algorithms, segment trees, combinatorics

Experience

Machine Learning Intern **Feb 2026 – Jul 2026**
Moleculyst *Remote*

- Designed and evaluated multimodal Video LLM inference pipelines — integrating vision-language architectures with structured output parsing, systematic benchmarking, and LLM-as-judge evaluation loops to assess model reasoning quality and output coherence across variants.
- Researched LLM architectures, AI agent frameworks, and vision-language models through hands-on experimentation, building practical understanding of how large models reason, retain context, and respond to tool-use and multi-turn instruction patterns.
- Co-built two internal AI products: **Agent Crop** — an agentic image-editing agent using VLMs and automated model critique loops — and **Moleculyst Hub** — an AI research-paper discovery and curation platform.
- Deepened mathematical foundations in linear algebra, probability theory, and optimization underlying modern LLM training, fine-tuning, and evaluation methodologies.

Projects

PaperLens | *Python 3.12, FastAPI, asyncio, Supabase, Multi-LLM* **2026**

- Built a 3-agent parallel paper explainer with auto-failover across 4 LLM providers (OpenAI, Groq, Cerebras, OpenRouter). Agents handle walkthroughs, architecture, and math analysis concurrently. Warm response in 10–15 s.
- Global cache eliminates repeat API costs. Shipped Research Radar with 1,100+ live items and 85 passing backend tests.

Equitas AI | *Python, LangGraph, FastAPI, SHAP, Gemini API, GCP* **2025**

- Built a 5-agent LangGraph pipeline for AI bias detection; achieved Disparate Impact Ratio $\geq$ **0.80** across all datasets.
- Combined SHAP explainability with Gemini-driven counterfactual narration. Deployed serverless on GCP Cloud Run via Vertex AI with Dockerised FastAPI.

UORA | *Python, FastAPI, Redis, TimescaleDB, Next.js, Docker* **2026**

- Built a distributed HFT benchmark platform with 4-layer decoupled architecture, deterministic LOBSTER replay, and L1–L4 mathematical correctness validation.
- Added ML anomaly detection via Isolation Forest on 8 features; real-time Next.js leaderboard with Redis Pub/Sub. 40 tests, full Docker Compose stack.

Triage AI | *Python, OpenEnv, LLaMA 3.1-8B, FastAPI, Docker* **2025**

- Built an RL environment that evaluates LLaMA 3.1 on email intent classification using structured OpenEnv state-machine cycles (reset / step / observe) with multi-level prompt engineering chains, achieving **0.71/1.0** reward accuracy.
- Containerised the full system with Docker and enforced API contracts with Pydantic, making it reproducible and easy to reset and re-evaluate across test cycles.

Achievements

- **Meta × PyTorch OpenEnv Hackathon — Grand Finalist** (Bangalore, 2025): Top-tier ranking out of **52,000+** global participants for shipping a production-ready RL environment.
- **SRMC 2025 — AIR 29 in Finals** (IIT Bombay & IIT Madras) | **SRMC 2026 — AIR 21 in Prelims**, Qualified for Finals (September 2026).
- **JEE Main 2024 — 98.16 percentile** (Top 1%, 1.2M+ candidates) | **JEE Advanced 2024 — Qualified**.
- **CodeBidz Hackathon — 1st Place** (Solo Developer) | **OpenAI × Outskill — Top 1,000** nationwide.